# The Laboratory Cannot Read This Patient

*A bovine oxygen carrier turns plasma the color of port wine, and every instrument built to distrust that color keeps right on distrusting it, which is a quieter hazard than any toxicity table admits*

## The tube that looks like an accusation

Picture a composite scene rather than a particular hospital. A patient who has lost most of a normal blood volume has been given a Biologic Hemoglobin Oxygen Carrier, a BHOC, because red cells were refused, unavailable or incompatible. Forty minutes later a phlebotomist draws a sample, spins it, and finds that the straw-yellow plasma that every technician has seen ten thousand times has turned a deep burgundy, a shade somewhere between cranberry juice and a cheap merlot. Nothing about this is a mistake. The carrier is hemoglobin, hemoglobin is red, and the carrier now travels dissolved in the plasma instead of folded inside a cell. But the laboratory does not know that, and the laboratory has been trained, by its instruments, its quality manuals and a century of accumulated habit, to read that color as a verdict against the person who drew the blood.

This essay is about that collision. Most writing on oxygen carriers asks whether the molecule works in the patient, whether it injures the vessel wall, whether it can be made at scale. Those questions deserve every page they receive. But a molecule that works in the body can still fail in the hospital if the hospital cannot see the patient through it, and the laboratory is the hospital's principal pair of eyes. I think the problem is under-discussed because it is unglamorous: it lives in middleware rules, wavelength tables and the small print of analyzer inserts, none of which appear in the press releases and few of which appear in the review articles.

## Why a laboratory is built to distrust free hemoglobin

To understand the collision you have to understand what the red color means to a chemistry analyzer. In ordinary life, hemoglobin in plasma almost always means that red cells broke open somewhere they should not have, usually in the tube, through a needle too narrow, a draw too forceful or a delay before spinning. The laboratory treats that as contamination, because breaking a red cell releases its contents, present at far higher concentrations inside the cell than outside. A veterinary pathology teaching site I consulted lists the standard consequences: falsely raised potassium, AST, LDH and magnesium, plus spectral effects on any test that reads light near the absorbance range of hemoglobin.

Modern analyzers measure this rather than eyeballing it. A laboratory-medicine primer on serum indices describes instruments reading the sample at several wavelengths, with hemoglobin absorbing light between roughly 340 and 440 nanometers and again between roughly 540 and 580, and computing a hemolysis index from wavelength pairs, 600 and 570 nanometers in the example of one widely used analyzer. Manufacturers publish limits for each assay, and laboratories write middleware rules, small programs between the analyzer and the hospital record, that stop results from being released automatically when the index exceeds a

limit. In the ordinary case the lab then asks for a fresh draw, hunts for an unspoiled specimen from the same collection time, or releases the number with a comment.

Now put a BHOC into this machinery. The same primer treats hemoglobin as a spectral and chemical nuisance. A carrier delivers hemoglobin on purpose, in concentrations that are not accidents of technique but a dose chosen by a prescriber. The analyzer sees a hemolysis index that would embarrass any phlebotomist, applies the rule it was told to apply, and withholds, flags or corrupts the result. A clinical laboratory page from a large academic center says the problem in one sentence that I find unusually clear: the presence of the carrier causes red-colored plasma that should not be confused with hemolysis of the red blood cells, and, strikingly, hemolysis cannot be detected in the presence of the carrier. I will return to that second clause, because I think it is the most consequential line in the entire literature on this subject.

## What the measurements show

The detail matters here, because the lazy summary, that these carriers wreck lab tests, is wrong in an instructive way. The truth is patchier and therefore harder to manage.

Consider the largest set of numbers I could verify. A study from 2018 tested the bovine carrier sold as Hemopure on four common chemistry platforms, covering 26 analytes across 104 combinations of assay, method and platform. The authors graded each result against the allowable total error for that test. In 44 of the 104 combinations, about 42 percent, the bias stayed within half the allowable error, which is the zone you can live with. In 23 combinations, about 22 percent, the carrier pushed the result significantly downward. In 26 combinations, 25 percent, it pushed the result significantly upward. In 11 combinations, about 11 percent, the analysis simply failed. Put differently, fewer than half of the tests you might order were trustworthy, and more than half either lied in one direction or the other or refused to answer. The authors attributed this to the extracellular hemoglobin mimicking severe hemolysis.

An earlier study, published in 1997 in Clinical Chemistry, took a more varied approach and tested several hemoglobin-based carriers alongside a perfluorocarbon emulsion. At concentrations up to 50 grams per liter, the hemoglobin carriers caused essentially no interference with electrolytes, glucose and standard immunoassays, while they significantly affected protein, enzyme and cholesterol measurements on certain analyzers. In coagulation testing the effect depended on method, with some techniques unaffected and others disturbed at 20 grams per liter. The cardiac marker CK-MB read falsely low at only 5 grams per liter. Among drug levels, one aminoglycoside antibiotic read falsely high and the glycopeptide vancomycin read falsely low. The perfluorocarbon, diluted twenty-fold, interfered with none of the chemistry or immunoassay tests studied except amylase, ammonia and plasma iron. I take two lessons from that older paper. First, the problem is test-specific and method-specific, so a one-page warning sheet will be wrong in half its lines. Second, the problem is physical rather than moral: a carrier built on a different principle, oxygen dissolved in fluorocarbon droplets, goes mostly unseen by the instruments, because it is not a colored protein.

Coagulation deserves a separate mention, because the workaround there is unusually clean. The academic laboratory page I consulted says that the prothrombin time and the activated partial thromboplastin time are disturbed when the instrument detects clotting optically, yet

can be measured accurately with mechanical, magnetic or light-scattering techniques, and that platelet counts can also be measured accurately. A patient on a carrier can therefore be monitored for bleeding, but the laboratory must route those samples to a particular instrument, and somebody must know to do that before a surgeon asks why the clotting numbers are absurd.

The pulse oximeter is the most reassuring case, and I include it for balance. A 1996 study of a bovine carrier compared finger-probe saturation with saturation from arterial blood gas analysis. Over 24 hours the mean difference in treated subjects was about 1.1 percent, with a standard error of 0.75, against 0.1 percent in controls, statistically significant but small, and the authors concluded that pulse oximetry stayed accurate across the normal range with plasma hemoglobin at 10 to 20 grams per liter. The laboratory page, by contrast, warns that the right-shifted dissociation curve may cause falsely high pulse readings and advises true saturation by co-oximetry. A shift of about one percentage point is small at the bedside and large in a research dataset, and the gap between what a clinician needs and what a trialist needs is part of the story.

## A single bed, closely read

Numbers across assays are abstractions. A case report gives the problem a pulse. The report, from a journal of blood research and disorders, describes a patient who was a Jehovah's Witness with life-threatening anemia and received the bovine carrier. I want to be careful here: it is one patient, reported by the team that treated her or him, and nothing about a single case generalizes. What it does show is how the laboratory problem looks from the bedside.

The authors write that the product is purple and causes errors in laboratory values, often being read as hemolyzed when it is not. They explain that the oxygen-carrying capacity has to be followed as total free hemoglobin, and that each unit raised free hemoglobin by about 0.63 grams per deciliter, although quantifying the response was difficult. Their pre-operative hemoglobin was 5 grams per deciliter, 5.7 by the sixth post-operative day and 9 by the sixteenth. Those figures describe a number on a report, not oxygen delivery. What they leaned on instead was the patient's behavior: they say they saw an effect immediately and weaned vasopressors within a day of surgery. The principal laboratory complication they followed was methemoglobin, hemoglobin with its iron oxidized and unable to carry oxygen. It was 4.6 percent on the first post-operative day, peaked at 8.6 percent early on the second, and returned to normal by the seventh. They gave vitamin C, 500 milligrams every twelve hours, and kept arterial oxygen pressures supratherapeutic at 300 to 400 millimeters of mercury.

Notice what is happening in that account. The team could not trust the ordinary hemoglobin number to mean what it normally means, so they substituted a different measurement and leaned on bedside physiology. A general principle, not something the report states, explains why. A routine hemoglobin measurement breaks open every red cell in the sample and reads all the hemoglobin that comes out, so it cannot distinguish hemoglobin that was inside cells from hemoglobin already floating free. A patient receiving a carrier therefore has a total figure that blends two very different oxygen-carrying systems, one confined to cells, the other dissolved and disappearing on a clock of its own. A 2023 study evaluated the bovine carrier on four hematology analyzers, but the journal page declined my attempts to read it, so I cannot report what it found, and I would rather leave that gap visible than fill it with a guess.

## The canary that goes silent

Return now to the sentence I flagged: hemolysis cannot be detected in the presence of the carrier.

In transfusion medicine, free hemoglobin in the plasma is one of the oldest alarms in the building. When a patient receives red cells of the wrong type, the recipient's antibodies destroy the donor cells, hemoglobin spills into plasma, and the plasma and urine change color. A technician who sees pink plasma after a transfusion reaction is trained to stop and think. That alarm exists because nature made red cells fragile in a specific way, and because plasma is supposed to be nearly colorless. A BHOC removes the second condition. If the plasma is already burgundy by design, the alarm has no sensitivity left. A real hemolytic event, whether from a mismatched red cell unit given in the same resuscitation, from a failing circulation, or from something the carrier itself did, would be hidden in the very place where it normally announces itself.

That is my reasoning rather than a documented incident. I did not find a case in the sources I read where a missed hemolytic event was traced to this mask, and I would be inventing a story if I suggested otherwise. But the structure is clear enough to name as a design risk. The settings where a carrier makes sense, massive hemorrhage, refusal of transfusion, remote injury, are the same settings where several fluids and drugs arrive at once, and the laboratory tools that would help sort out which of them did what are the ones the carrier has degraded.

I want to push a second consequence, because I rarely see it discussed. A red tube is also a break in blinding. If a randomized trial compares a carrier to a standard fluid, every phlebotomist, technician and nurse can see which arm a patient is in by looking at a centrifuged specimen. Allocation concealment may survive at the pharmacy door and fail at the sample rack. The concern is observer bias, sharpest for judgment-dependent outcomes such as whether to transfuse a rescue unit or repeat a test. Independent adjudicators who see only coded data, and transfusion triggers fixed in advance, are the known defenses, and the physical color of the specimen is a reason to build them on purpose.

## Workarounds that already exist

The encouraging news is that laboratory medicine has solved harder interference problems than this, and the tools exist. They are scattered, and they are rarely collected in one place for the people who need them.

For clotting, the mechanical, magnetic and light-scattering methods avoid the optical trap. For oxygen saturation, co-oximetry, which reads many wavelengths and measures several hemoglobin species, gives a truer answer than a two-wavelength finger probe, although a literature exists showing that hemoglobin-based substitutes can disturb co-oximetry too, so I would not treat any single instrument as final authority. The same method tracks methemoglobin, as the case report shows. For total hemoglobin, plasma hemoglobin can be measured separately, revealing how much capacity sits in each compartment. For the chemistry panel, the 2018 numbers imply a pragmatic rule: assume nothing, and validate each test on the site's own instruments with spiked samples before anyone trusts it.

These workarounds all presume that a laboratory knows a patient has received a carrier, and in the settings that matter most that is not guaranteed. A patient moved from a roadside to a regional trauma center to a referral hospital may pass through three laboratories, and the news that burgundy plasma is intentional lives, at best, in a handoff note. Middleware that auto-rejects hemolyzed samples can do so without a human looking, and the requesting physician may see only the word hemolyzed. The system fails quietly, which is the worst way to fail.

## What a laboratory passport would contain

These are my own suggestions, drawn from the structure of the problem and not tested anywhere I know of. The first element would be a flag in the laboratory information system, set at the time of administration, that travels with the patient record and tells the lab to treat red plasma as expected. A middleware rule that suppresses automatic rejection for flagged patients is cheap compared with a repeat draw from a patient who has little blood to spare.

The second would be a validated test menu, built before deployment. For each analyzer a site uses, a laboratory scientist would spike donor plasma with the carrier across the concentrations expected after a dose, run the routine tests, and record whether each can be trusted, which way it errs, and which alternative method to use. The 2018 study shows why this must be done per platform: 26 analytes and four instruments made 104 situations, and fewer than half behaved.

The third would be a physical marker: a wristband or chart sticker saying that the patient has received a bovine oxygen carrier and that plasma color is expected, with a telephone number. It is a low-technology answer to a high-technology failure, and a laboratory worker who sees the burgundy tube will instinctively look for an explanation.

The fourth would be a written specification of which measurements a trial will use for its endpoints, chosen before enrollment, with methods that survive the carrier. If a protocol depends on a number that the product itself distorts, the trial has built a hole into its own result.

The fifth, and the one I consider most important, is shared learning. The papers I found are scattered across clinical chemistry, anesthesia, hematology, coagulation, veterinary pathology and case reports, and one page I consulted says that these products are no longer used, a sentence that shows how easily the knowledge could be forgotten while the products remain in development. Interference data should be a standing appendix to any serious BHOC dossier, updated as instruments change.

## Why I trust the problem more than the product

There is a quieter lesson as well, and here the labyrinthine detail resolves into a single point. The history of this field is usually told as a history of toxicity, with vasoconstriction and heart injury receiving most of the attention. Those are real. But a molecule can also fail by being unmeasurable. It is not a death sentence: electrolytes and glucose were essentially untouched in the older study, and workarounds exist for clotting and saturation. Nor is it a footnote: more than half the assay combinations in the best dataset I could verify were biased or failed. If nobody can reliably say whether a patient is improving, nobody can reliably say whether a

treatment works, and an unmeasurable benefit is, for regulatory and clinical purposes, nearly the same as none. The laboratory is not downstream of the science. It is part of it.

I do not know how long burgundy plasma persists after a typical dose, because that depends on dose and clearance, and I found no figure I trust enough to print. I also do not know how current analyzers behave, because most of the literature concerns instruments that were current when the studies ran, some of them decades ago. Those uncertainties are themselves a finding, since a field that wishes to be taken seriously should be able to say, in a table, how its product interacts with the instruments of every hospital that might receive it.

## Further reading

For the quantitative picture, look for the 2018 study of the bovine carrier Hemopure on four clinical chemistry platforms, and the 1997 Clinical Chemistry paper on hemoglobin and perfluorocarbon carriers and common laboratory tests. For saturation, the 1996 study comparing pulse oximetry with arterial blood gas saturation after a bovine carrier, and a 2001 Anesthesia and Analgesia paper on co-oximetry interference, whose full text I could not open. For coagulation, a 2001 paper on the bovine carrier and coagulation testing, which I could not retrieve and cite only by subject. For the laboratory side, the Clinical Laboratory News primer on detecting and handling hemolysis using serum indices, and the University of California San Francisco clinical laboratory page on blood substitutes. For the bedside, the case report in the International Journal of Blood Research and Disorders. Figures here are quoted as I read them in abstracts or summaries; check the originals before relying on them.